

Polymetallic Mineralization at the Suttu Vein-type Deposit, Southwestern Hokkaido, Japan

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Abstract: The Suttu polymetallic vein-type deposit, hosted by tuff, tuff breccia and shale of the Miocene Kunnui Formation and propylitized hornblende-augite andesite, is located in southwestern Hokkaido, Japan. It has been exploited and explored for Cu, Pb, Zn and Ag until 1962. In this study, we examined K-Ar ages, ore mineralogical characteristics and fluid inclusions to obtain new data for the deposit.

The K-Ar ages on sericite indicate that the polymetallic mineralization occurred in Late Miocene (8.1–5.7 Ma). The polymetallic banded ore from the Ohkubo vein is characterized by an abundance of Au, Ag, Sn, Bi, In, Se and Te. These metals are mainly ascribed to electrum (30.3–37.8 atom% Ag), Se-bearing pavonite (8.5–9.5 wt% Se), gustavite-lillianite solid solution, Se-bearing bismuthinite (5.0–5.3 wt% Se), kawazulite, cassiterite, Sn-bearing chalcocopyrite (3.3–4.2 wt% Sn), In-bearing stannite, stannite-chalcocopyrite solid solution, and In- and Sn-bearing sphalerite (2.6–8.4 wt% In and 1.8–4.3 wt% Sn), occurring in narrow bands of the ore. The In- and Sn-bearing sphalerite likely forms a sphalerite-roquesite-stannite solid solution with the contents of roquesite and stannite being about 2–9 and 2–4 mole%, respectively. Temperatures and salinities (in wt% NaCl equiv.) of the ore fluids are estimated to be 180°–250°C and 3–4 wt%, respectively. The Sn-bearing chalcocopyrite therefore probably precipitated metastably. The geologic and mineralogical features suggest that pre-Tertiary basement rocks rich in organic material underlie the Miocene Kunnui Formation nearby the deposit and that they contributed to local and temporary reduction of magnetite-series magmas favorable for the early stage tin-polymetallic mineralization.

Keywords: polymetallic mineralization, Suttu, vein-type, silver, tin, bismuth, indium, selenium, tellurium, stannite-chalcocopyrite solid solution

1. Introduction

The Suttu polymetallic vein-type deposit, located in southwestern Hokkaido, Japan (Fig. 1), was discovered in 1892 and has been exploited and explored intermittently for Cu, Pb, Zn and Ag until 1962. The total production of crude ores from the mine is about 196,000 metric tons from 1943 to 1962 (Geological Survey of Japan, 1967).

Published studies on geology and ore petrography of the deposit have been very scarce, except for Ishibashi (1952). On the basis of ore microscopic observation, he noticed the Sn-Bi-Sb-Te mineralization characterized by the occurrence of cassiterite, stannite, tetradymite, bismuthinite and famatinite in the deposit. A mineralogical study has recently been made by Shimizu et al. (1995), who identified Bi-Te minerals from Suttu as kawazulite, selenian bismuthinite, selenian pavonite and an unnamed mineral with the ideal formula $\text{Bi}_3(\text{Te}, \text{Se}, \text{S})_4$ by means of microprobe analysis. In addition, mohite occurs with stannoidite, colusite and nekrasovite, associated with Au mineralization in the deposit (Masaaki Shimizu, Unpubl. data).

In this paper, we report the results of K-Ar age determination, modes of occurrence and chemistries of ore miner-

als, and fluid inclusion microthermometric data for the Suttu deposit. The ore samples examined in this study were collected by the late Professor Masao Ishibashi of Hokkaido University.

2. Geology and Ore Deposit

The descriptions in this chapter are summarized mainly from Ishibashi (1952).

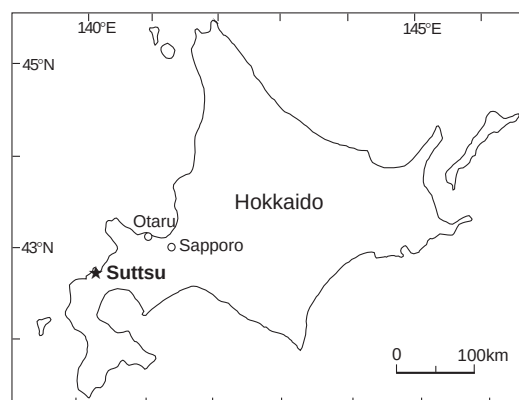


Fig. 1 Location map of the Suttu deposit, southwestern Hokkaido, Japan.

2.1. Geology

The Suttu mining area is composed of the Miocene Kunnui Formation (green or greenish white tuff, tuff breccia, hyaloclastite and shale), Pliocene Kuromatsunai Formation (andesitic volcanic flow and pyroclastics) and Pleistocene marine terrace deposit. The Kunnui Formation, which hosts the ore veins, is intruded by many small dikes of propylitized hornblende-augite andesite and augite andesite. The former dikes seem to have a close genetic relationship with mineralization, since some pyrite-quartz veins and pyrite impregnated low grade ores are often found in the green tuff adjacent to those dikes and in the dikes themselves. In the area about 1 km north-west of the deposit, a large intrusive body of quartz porphyry is exposed and an age of the intrusion is inferred to be Late Miocene (Suzuki et al., 1981). Although the quartz porphyry is weakly propylitized, it shows a high magnetic susceptibility.

2.2. Vein system

In the Suttu mine, there occur the Honpi vein (mother lode) and several small veins. The Honpi vein strikes EW and dips 90–70°S, with strike extension of about 350 m and dip extension of over 230 m. It generally consists of two or three thin lenticular veins, 30 to 50 cm in thickness, that separate and adjoin. The NE-striking Hokutoh vein branches off at the middle part of the Honpi vein. The Ohkubo vein, running parallel to the Honpi vein, dips 80°N. The host rocks, including tuff, tuff breccia and shale close to these veins, have commonly undergone silicification and kaolinization. Chloritization and pyritization have developed a little more widely in the host rocks.

The Honpi vein, showing massive or banded structure, consists mainly of pyrite, sphalerite, galena and quartz. Cassiterite is also found in places. The brecciated ore in which fragments of pyrite or sphalerite are cemented by quartz or Mn-calcite is rarely found in the vein. The structure and mineral composition of the Ohkubo vein are similar to those of the Honpi vein. Pyrite, gypsum, barite or calcite crystals covered with clay minerals often occur in vugs of these veins. In the western portion on the No. 5 sublevel (–35 m in elevation) of the Ohkubo vein, siliceous vein with Sn-Bi-Te-Sb-(Cu) mineralization occurs characteristically (Fig. 2), and it generally shows banded structure with pyrite, galena, sphalerite and chalcopryrite. The vein structure suggests that the siliceous Sn-Bi-Te-Sb-(Cu) mineralization predated the FeS₂-Pb-Zn-Cu one.

3. K-Ar Age

Two sericites separated by hydraulic elutriation from ore and altered tuff breccia samples were dated by K-Ar method at the Geochron Laboratory of the Kruger Enterprises, Inc., U. S. A. The ore sample (STT-06) collected from the Hokutoh vein (No. 15 level (–190 m in

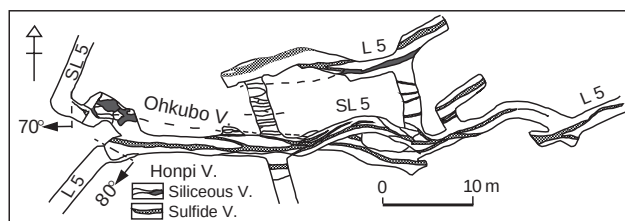


Fig. 2 Vein map of No. 5 sublevel (–35 m level) and No. 5 level (–40 m level) (modified after Ishibashi, 1952).

Table1 K-Ar ages of sericite from the Suttu deposit.

Sample No.	Sample analyzed	Mineral analyzed	K (wt%)	Rad. Ar (ppm)	Air cont. (%)	K-Ar age (Ma)
STT-006	Ore	Sericite	1.250	0.000666	97.0	7.7±0.4
STT-001	Altered tuff breccia	Sericite	5.431	0.002206	83.7	5.9±0.2

Analyst : Geochron Lab., Krueger Enterprises Inc.

Constants used: $\lambda_p = 4.962 \times 10^{-10}/y$ and $\lambda_e = 0.581 \times 10^{-10}/y$.

$^{40}K/K = 1.193 \times 10^{-4}$ g/g

elevation), E 1) consists mostly of coarse-grained sphalerite and galena coated with sericite. The tuff breccia (STT-01), having undergone strong hydrothermal silicification, sericitization and pyritization, is in contact with a banded ore composed mainly of quartz, cassiterite, pyrite, sphalerite, galena, chalcopryrite, stannite, kawazulite and marcasite pseudomorphs after pyrrhotite (Fig. 3A). This sample is from the Ohkubo vein (No. 5 level (–40 m in elevation), W 90 m). The sericite concentrates were analyzed with an X-ray powder diffractometer (XRD) before the age determination. XRD analyses detected trace amounts of kaolinite in the two concentrates. The d (001) of sericite from the ore is 10.1 Å, whereas that from the tuff breccia is 10.4–10.5 Å. A little swelling was observed by ethylene glycol treatment in the latter sericite.

The K-Ar ages obtained from the ore and tuff breccia are 7.7±0.4 and 5.9±0.2 Ma, respectively (Table 1). The latter is identical to the age of sericite in strongly altered andesitic tuff (5.9±0.3 Ma; Ministry of Economy, Trade and Industry (METI), 2001) collected from about 4 km south-southeast of the Suttu deposit (42°45'20"N, 140°14'44"E).

4. Ore Petrography and Mineral Chemistry

4.1. Samples and analytical methods

Ore microscopic observation was carried out on two ore samples: (1) Pb-Zn-Cu ore (STT-13) collected from the Honpi vein (No. 8 level (–85 m in elevation)), and (2) polymetallic banded ore (STT-03) from the Ohkubo vein (No. 5 sublevel (–35 m in elevation), W 110 m).

Qualitative and quantitative chemical analyses of ore minerals from the polymetallic banded ore were performed using a JEOL 8900M electron probe microanalyzer. Back-scattered electron and characteristic X-ray images were also examined for stannite-chalcopryrite

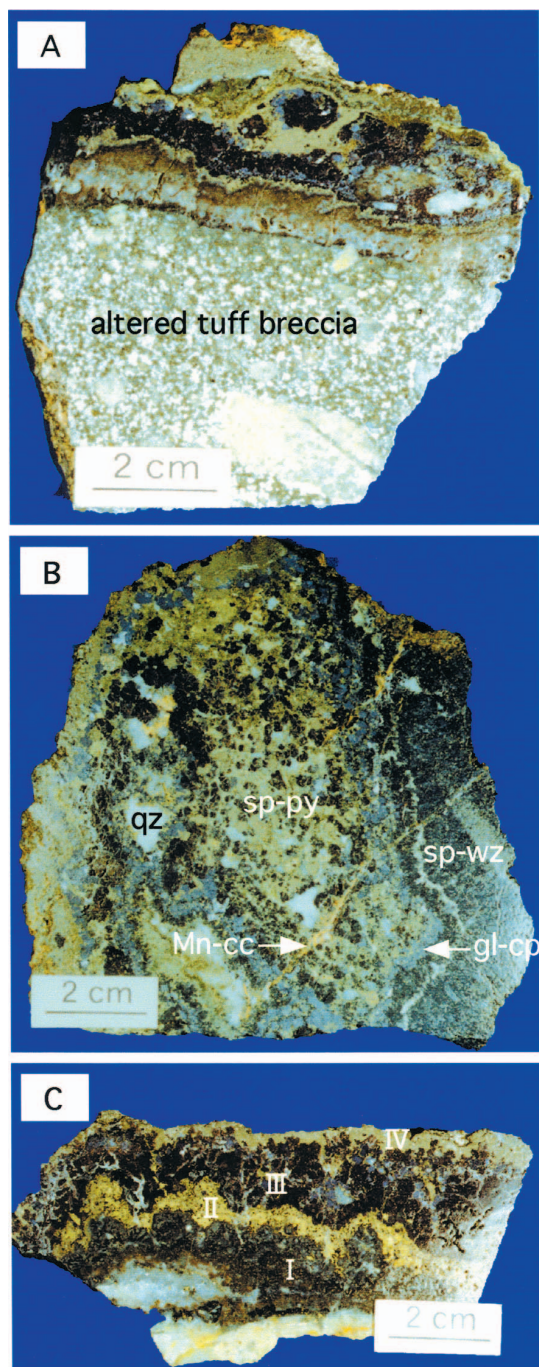


Fig. 3. Photographs of ore samples examined in this study. A. Hydrothermally altered tuff breccia used for K-Ar age determination (STT-01) (Ohkubo vein, No. 5 level (–40 m), W 90 m). B. Pb-Zn-Cu ore (STT-13) (Honpi vein, No. 8 level (–85 m)). sp: sphalerite, py: pyrite, gl: galena, cp: chalcopyrite, wz: wurtzite, qz: quartz, Mn-cc: Mn-calcite. C. Polymetallic banded ore (STT-03) (Ohkubo vein, No. 5 sublevel (–35 m), W 110 m). Characteristic ore minerals in the zones I to IV are as follows: (I) cassiterite, kawazulite, bismuthinite, pavonite, gustavite-lillianite solid solution, stannite-chalcopyrite solid solution and electrum; (II) chalcopyrite, stannite-chalcopyrite solid solution and In-bearing stannite; (III) normal and In- and Sn-bearing sphalerite; and (IV) pyrite.

solid solution and In- and Sn-bearing sphalerite in the ore. Operating conditions for quantitative analysis were: an accelerating voltage of 20 kV and a probe current of 20 nA. The elements quantitatively analyzed were Au, Ag, Cu, Fe, Pb, Cd, Mn, As, Bi, In, Sn, Se, Te and S. Standards used were chalcopyrite for Cu, Fe and S, galena for Pb, synthetic In_2Se_3 for In and pure metals for other elements. The corrections of count data were made by Bence-Albee method for electrum and ZAF method programmed by JEOL for other minerals.

4.2. Bulk metal contents

Bulk metal contents of the ores (METI, 2001, 2002) are given in Table 2. The Au, Ag, Sn, Bi, Te and Se contents of the polymetallic banded ore (STT-03) are remarkably high compared with those of the Pb-Zn-Cu ore (STT-13). Moreover, the In content of the polymetallic banded ore is also significantly high (290 ppm).

4.3. Pb-Zn-Cu ore

The Pb-Zn-Cu ore consists of coarse-grained sphalerite-pyrite aggregates, and galena-chalcopyrite and sphalerite-wurtzite bands (Fig. 3B). The aggregates and bands are cut by Mn-calcite veinlet. The sphalerite-pyrite aggregates also contain subordinate amounts of quartz, small amounts of galena and chalcopyrite, and trace amounts of tetrahedrite-tennantite. The galena-chalcopyrite band includes pyrite, sphalerite and quartz as accessory minerals. The sphalerite-wurtzite band contains subordinate amounts of galena, pyrite, chalcopyrite and quartz. Under the microscope with transmitted light, wurtzite exhibits a fibrous texture about 0.2 to 2 mm long.

4.4. Polymetallic banded ore

The polymetallic banded ore comprises discrete bands of quartz and zones I to IV composed of different mineral assemblages (Fig. 3C). The bands of quartz and these four zones formed in this order.

4.4.1. Zone I: The zone I, ranging in width from 1 to 2 cm, is characterized by the dominance of cassiterite and Bi-minerals such as kawazulite, bismuthinite, pavonite and gustavite-lillianite solid solution in quartz matrix. The high concentrations of Ag, Sn, Bi, Te and Se in the ore are ascribable to these minerals. Other ore minerals observed in this zone are chalcopyrite, pyrite and stannite-chalcopyrite solid solution in subordinate amounts and some grains of electrum.

Cassiterite, ranging from about 10 to 50 μm in size, abundantly occurs as idiomorphic and hypidiomorphic crystals in quartz (Fig. 4A). The grains associated with Bi-minerals are also common.

Kawazulite, the most abundant Bi-mineral and generally $\leq 50 \mu\text{m}$, often occurs between grains of quartz (Fig. 4A). Under the microscope, kawazulite is white with a creamy tint, with weak pleochroism and distinct anisotropy.

Table 2 Chemical analyses of bulk ore samples (METI, 2001, 2002).

Sample No.	Au (ppm)	Ag (ppm)	Hg (ppb)	Cu (%)	Pb (%)	Zn (%)	Fe (%)	Sn (ppm)	Bi (ppm)	In (ppm)	As (ppm)	Sb (ppm)	Te (ppm)	Se (ppm)
STT-03	5.52	379	20	3.17	1.95	6.75	10.26	43,000	35,600	290	259	115	16,100	5,360
STT-13	1.21	95.0	20	0.52	9.81	28.20	10.82	289	74	n.a.	578	104	31	11

n. a. = not analyzed.

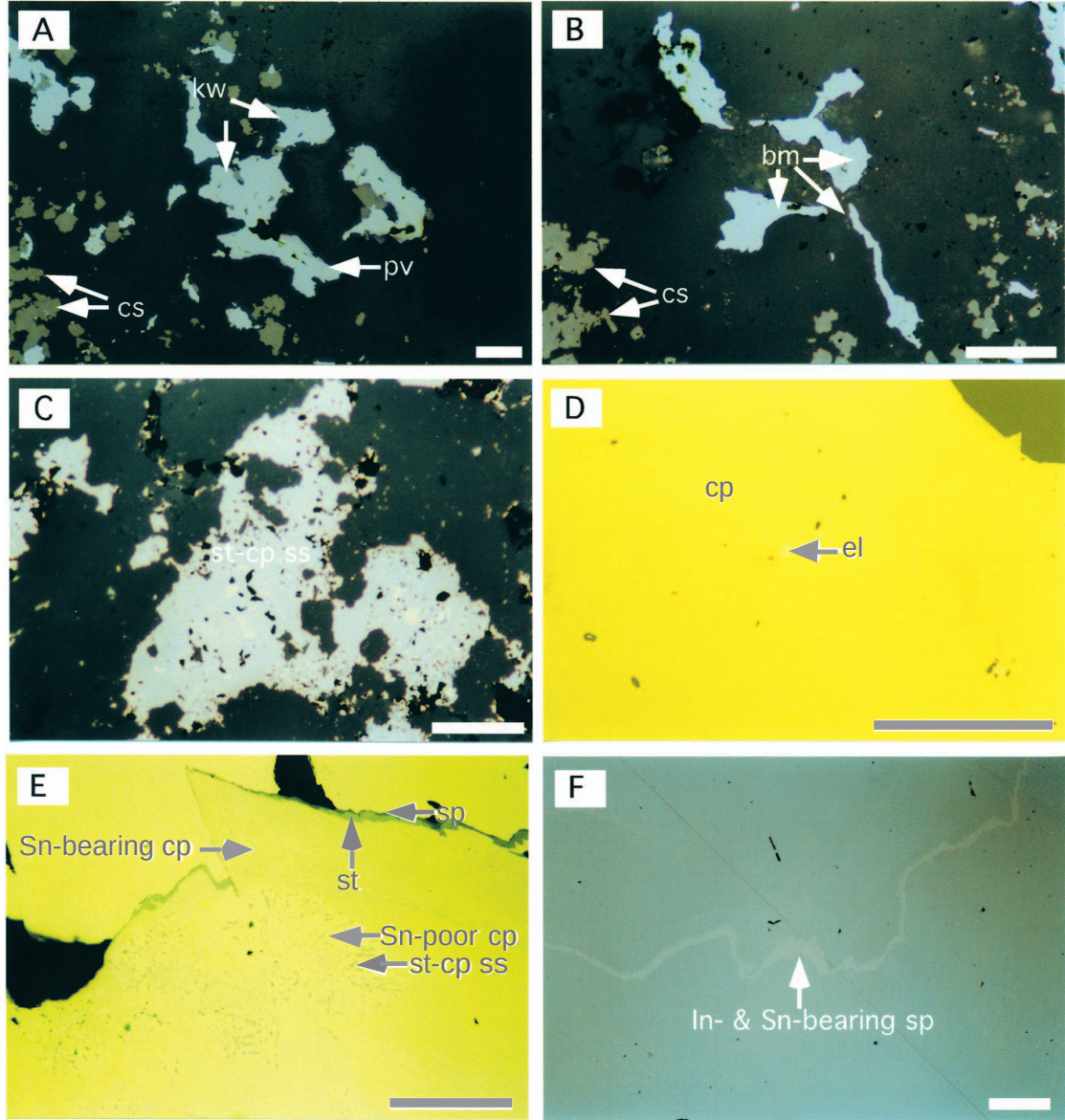


Fig. 4 Photomicrographs of polymetallic banded ore (STT-03). Scale bars are 50 μ m long. A. Cassiterite (cs), kawazulite (kw) and pavonite (pv) in quartz. B. Bismuthinite (bm) and cassiterite (cs) in quartz. C. Stannite-chalcopyrite solid solution (st-cp ss) in quartz. D. Electrum (el) in chalcopyrite (cp). E. Sn-bearing chalcopyrite (cp), Sn-poor cp (normal cp) and stannite-chalcopyrite solid solution (st-cp ss) rimmed by sphalerite (sp) and In-bearing stannite (st). F. In- and Sn-bearing sphalerite (sp) in normal sphalerite.

pism (dark gray with brownish and bluish tints). The S contents of kawazulite are fairly high (2.3–3.3 wt%) (Table 3) in comparison with the value (1.9 wt%) reported by Shimizu et al. (1995).

The mode of occurrence of bismuthinite is similar to that of kawazulite (Fig. 4B). Bismuthinite contains about

5 wt% of Se, and minor amounts of Cu (0.1–0.2 wt%) and Pb (0.4–0.6 wt%) (Table 3).

Pavonite occurs as aggregates of crystals up to 0.1 mm in size (Fig. 4A). Under the microscope, the reflection pleochroism is distinct, white to light gray with a pinkish tint, and the reflectance is slightly lower than that of

kawazulite, with strong anisotropism (intense blue to brown). The pavonite contains considerable Se (8.5–9.5 wt%) and minor amounts of Cu (0.3 wt%), Pb (3.9–4.1 wt%) and Te (0.3–0.5 wt%). The empirical formula nearly corresponds to (Ag, Cu) (Bi, Pb)₃(S, Se, Te)₅ on the basis of nine total number of atoms (Table 4).

Gustavite-lillianite solid solution is found as aggregates of crystals up to 0.1 mm in size. It is often associated with chalcopyrite in quartz. Under the microscope, it shows distinct pleochroism (white to light gray) and strong anisotropism (bluish gray to brownish gray). Gustavite-lillianite solid solution contains minor amounts of Se (0.4–1.4 wt%) and Te (0.1–0.2 wt%). The numbers of Pb, Ag, Bi and [S+Se+Te] on the basis of eleven total atoms are 1.32–1.39, 0.75–0.79, 2.81–2.86 and 6.00–6.04, respectively (Table 4).

Stannite-chalcopyrite solid solution occurs as aggregates of irregular grains up to 50 µm across (Fig. 4C). The aggregates always involve minute dots (<2 µm) and small grains (5–15 µm) of chalcopyrite. The aggregates commonly consist of two types of grains, which have different optical properties: (1) strong pleochroism (gray with an olive green tint to brownish gray) and anisotropism (intense blue to yellowish brown), and (2) weak to moderate pleochroism and anisotropism. As shown in Table 5, electron microprobe analyses indicate that the former grains (Nos. 1 and 2) are poor in Sn and rich in (Fe+Zn) compared with the latter ones (Nos. 3–5). This implies that the chalcopyrite mole fraction in the solid solution is distinctly higher in the strongly anisotropic grains. In addition, the grains show a

Table 3 Electron microprobe analyses of kawazulite and bismuthinite.

Mineral		Ag	Cu	Pb	Bi	Te	Se	S	Total
kawazulite	wt%	0.05	0.06	0.04	60.12	29.33	8.54	2.99	101.13
	atom%	0.06	0.13	0.03	39.93	31.90	15.02	12.94	100.01
	no. of atoms	0.003	0.006	0.001	1.996	1.595	0.751	0.647	5
kawazulite	wt%	0.06	0.00	0.00	61.40	25.67	10.27	3.25	100.65
	atom%	0.07	0.00	0.00	40.41	27.68	17.89	13.95	100.00
	no. of atoms	0.004	0.000	0.000	2.021	1.384	0.894	0.698	5
kawazulite	wt%	0.03	0.00	0.00	60.29	24.82	11.48	2.99	99.61
	atom%	0.04	0.00	0.00	39.96	26.95	20.15	12.91	100.01
	no. of atoms	0.002	0.000	0.000	1.998	1.347	1.007	0.645	5
kawazulite	wt%	0.03	0.00	0.11	60.25	25.38	11.75	2.64	100.16
	atom%	0.04	0.00	0.07	40.10	27.66	20.69	11.44	100.00
	no. of atoms	0.002	0.000	0.004	2.005	1.383	1.034	0.572	5
kawazulite	wt%	0.07	0.00	0.00	59.87	28.82	8.61	2.97	100.34
	atom%	0.09	0.00	0.00	40.10	31.60	15.27	12.95	100.01
	no. of atoms	0.004	0.000	0.000	2.005	1.580	0.763	0.647	5
kawazulite	wt%	0.02	0.02	0.00	59.31	28.75	8.64	2.86	99.60
	atom%	0.03	0.05	0.00	40.06	31.80	15.45	12.60	99.99
	no. of atoms	0.000	0.002	0.000	2.003	1.590	0.773	0.630	5
kawazulite	wt%	0.07	0.01	0.00	59.39	27.92	9.92	2.33	99.64
	atom%	0.09	0.03	0.00	40.47	31.16	17.90	10.36	100.01
	no. of atoms	0.004	0.001	0.000	2.023	1.558	0.895	0.518	5
kawazulite	wt%	0.06	0.00	0.00	61.31	23.99	11.92	3.09	100.37
	atom%	0.07	0.00	0.00	40.23	25.79	20.70	13.21	100.00
	no. of atoms	0.004	0.000	0.000	2.012	1.289	1.035	0.660	5
bismuthinite	wt%	0.00	0.13	0.48	78.43	0.00	5.26	15.55	99.85
	atom%	0.00	0.21	0.25	40.30	0.00	7.15	52.09	100.00
	no. of atoms	0.000	0.011	0.012	2.015	0.000	0.357	2.605	5
bismuthinite	wt%	0.04	0.19	0.41	79.38	0.00	4.95	15.92	100.89
	atom%	0.04	0.31	0.21	40.22	0.00	6.63	52.57	99.98
	no. of atoms	0.002	0.016	0.011	2.011	0.000	0.332	2.629	5
bismuthinite	wt%	0.00	0.09	0.35	78.04	0.00	5.12	15.49	99.09
	atom%	0.00	0.15	0.18	40.39	0.00	7.02	52.25	99.99
	no. of atoms	0.000	0.008	0.009	2.020	0.000	0.351	2.613	5
bismuthinite	wt%	0.01	0.17	0.55	78.05	0.03	5.08	15.59	99.48
	atom%	0.01	0.29	0.29	40.17	0.02	6.92	52.30	100.00
	no. of atoms	0.001	0.015	0.014	2.009	0.001	0.346	2.615	5

Table 4 Electron microprobe analyses of pavonite and gustavite-lillianite solid solution.

Mineral		Ag	Cu	Pb	Bi	Te	Se	S	Total
pavonite	wt%	11.45	0.28	4.10	62.80	0.50	9.49	12.85	101.47
	atom%	11.10	0.45	2.07	31.45	0.41	12.58	41.93	99.99
	no. of atoms	0.999	0.041	0.186	2.830	0.037	1.132	3.774	9
pavonite	wt%	11.12	0.30	4.03	62.18	0.38	8.62	13.06	99.69
	atom%	10.92	0.50	2.06	31.51	0.32	11.57	43.14	100.02
	no. of atoms	0.983	0.045	0.185	2.836	0.028	1.041	3.883	9
pavonite	wt%	11.05	0.32	3.99	62.46	0.34	8.78	13.08	100.02
	atom%	10.81	0.53	2.03	31.54	0.28	11.74	43.07	100.00
	no. of atoms	0.973	0.047	0.183	2.839	0.025	1.056	3.876	9
pavonite	wt%	11.14	0.30	3.90	62.53	0.43	8.53	13.03	99.86
	atom%	10.94	0.49	2.00	31.70	0.36	11.45	43.05	99.99
	no. of atoms	0.985	0.044	0.180	2.853	0.032	1.030	3.875	9
gustavite-lillianite ss	wt%	7.22	0.01	24.35	51.63	0.10	1.40	16.01	100.72
	atom%	7.05	0.02	12.37	26.02	0.08	1.87	52.58	99.99
	no. of atoms	0.776	0.002	1.361	2.862	0.009	0.206	5.784	11
gustavite-lillianite ss	wt%	6.94	0.04	24.57	49.96	0.15	0.78	16.15	98.59
	atom%	6.86	0.06	12.65	25.51	0.12	1.06	53.74	100.00
	no. of atoms	0.754	0.007	1.392	2.806	0.013	0.116	5.911	11
gustavite-lillianite ss	wt%	7.37	0.32	23.69	51.06	0.08	0.36	16.55	99.43
	atom%	7.16	0.54	11.99	25.62	0.06	0.48	54.15	100.00
	no. of atoms	0.788	0.059	1.319	2.818	0.007	0.053	5.956	11

Table 5 Electron microprobe analyses of stannite-chalcopyrite solid solution series.

No.	Mineral		Cu	Ag	Fe	Zn	Mn	Cd	In	As	Sn	S	Total
1	st-cp ss	wt%	32.19	0.00	13.56	4.19	n. a.	n. a.	0.01	0.22	18.35	31.04	99.56
		atom%	26.12	0.00	12.52	3.31			0.00	0.15	7.97	49.93	100.00
		no. of atoms	2.090	0.000	1.002	0.265			0.000	0.012	0.638	3.994	8
2	ditto	wt%	32.53	0.05	13.41	4.42	n. a.	n. a.	0.03	0.13	18.53	30.93	100.03
		atom%	26.35	0.02	12.36	3.48			0.01	0.09	8.04	49.66	100.01
		no. of atoms	2.108	0.002	0.988	0.278			0.001	0.007	0.643	3.973	8
3	ditto	wt%	29.11	0.14	9.75	7.42	n. a.	n. a.	0.07	0.45	22.27	30.62	99.83
		atom%	24.15	0.07	9.20	5.98			0.03	0.32	9.89	50.35	99.99
		no. of atoms	1.932	0.006	0.736	0.479			0.003	0.025	0.791	4.028	8
4	ditto	wt%	29.93	0.00	9.00	4.98	n. a.	n. a.	0.06	0.02	26.36	29.83	100.18
		atom%	25.30	0.00	8.65	4.10			0.03	0.01	11.93	49.98	100.00
		no. of atoms	2.024	0.000	0.692	0.328			0.002	0.001	0.954	3.999	8
5	ditto	wt%	30.72	0.14	9.94	4.68	n. a.	n. a.	0.05	0.53	23.35	30.51	99.92
		atom%	25.57	0.07	9.41	3.79			0.02	0.38	10.41	50.35	100.00
		no. of atoms	2.046	0.006	0.753	0.303			0.002	0.030	0.833	4.028	8
6	Sn-bearing cp	wt%	33.57	n. a.	27.30	0.63	0.00	0.00	0.09	n. a.	3.63	34.21	99.43
		atom%	24.86		23.00	0.45	0.00	0.00	0.03		1.44	50.21	99.99
		no. of atoms	1.989		1.840	0.036	0.000	0.000	0.002		0.115	4.017	8
7	ditto	wt%	33.60	n. a.	27.47	0.51	0.01	0.00	0.06	n. a.	3.34	34.01	99.00
		atom%	24.96		23.22	0.37	0.01	0.00	0.02		1.33	50.09	100.00
		no. of atoms	1.997		1.858	0.030	0.001	0.001	0.002		0.106	4.007	8
8	ditto	wt%	33.60	n. a.	27.00	0.67	0.00	0.00	0.06	n. a.	4.22	34.00	99.55
		atom%	24.95		22.81	0.48	0.00	0.00	0.03		1.68	50.05	100.00
		no. of atoms	1.996		1.825	0.038	0.000	0.000	0.002		0.134	4.004	8
9	ditto	wt%	33.33	n. a.	26.90	0.76	0.00	0.00	0.07	n. a.	3.94	33.85	98.85
		atom%	24.89		22.86	0.55	0.00	0.00	0.03		1.57	50.10	100.00
		no. of atoms	1.991		1.829	0.044	0.000	0.000	0.002		0.126	4.008	8
10	st-cp ss (myrmekitic)	wt%	30.87	n. a.	17.22	3.18	0.04	0.00	0.26	n. a.	16.30	31.32	99.19
		atom%	24.79		15.74	2.48	0.03	0.00	0.11		7.01	49.84	100.00
		no. of atoms	1.983		1.259	0.198	0.002	0.000	0.009		0.561	3.987	8
11	ditto	wt%	30.05	n. a.	13.22	4.11	0.05	0.01	0.36	n. a.	21.49	30.44	99.73
		atom%	24.80		12.41	3.30	0.05	0.00	0.16		9.50	49.79	100.01
		no. of atoms	1.984		0.993	0.264	0.004	0.000	0.013		0.760	3.983	8
12	ditto	wt%	29.87	n. a.	13.12	4.14	0.05	0.00	0.27	n. a.	22.15	30.55	100.15
		atom%	24.59		12.30	3.32	0.04	0.00	0.12		9.77	49.86	100.00
		no. of atoms	1.967		0.984	0.266	0.003	0.000	0.010		0.782	3.989	8
13	cp (myrmekitic)	wt%	34.00	n. a.	29.51	0.13	0.00	0.02	0.01	n. a.	0.53	34.46	98.66
		atom%	24.94		24.63	0.09	0.00	0.01	0.01		0.21	50.11	100.00
		no. of atoms	1.995		1.970	0.007	0.000	0.001	0.001		0.017	4.009	8
14	ditto	wt%	33.77	n. a.	29.80	0.00	0.00	0.00	0.03	n. a.	0.12	34.76	98.48
		atom%	24.71		24.81	0.00	0.00	0.00	0.01		0.05	50.42	100.00
		no. of atoms	1.977		1.985	0.000	0.000	0.000	0.001		0.004	4.034	8
15	In-bearing st	wt%	27.40	n. a.	8.64	8.34	0.08	0.01	2.99	n. a.	22.68	29.64	99.78
		atom%	23.22		8.34	6.87	0.07	0.00	1.40		10.29	49.79	99.98
		no. of atoms	1.858		0.667	0.550	0.006	0.000	0.110		0.823	3.983	8

st = stannite, cp = chalcopyrite, ss = solid solution, n. a. = not analyzed.

slight excess of Cu content. This may be due to the dissolved kuramite ($\text{Cu}^{1+}_2\text{Cu}^{2+}\text{SnS}_4$) as a minor component. The In and As contents of the two types of grains are <0.1 wt% and 0.1–0.5 wt%, respectively.

Electrum is observed as minute inclusions (<10 μm in diameter) in chalcopyrite (Fig. 4D). The Ag contents of electrum range from 30.3 to 37.8 atom% (Table 6).

4.4.2. Zone II: The zone II, ranging in width from 0.2 to 1 cm, consists of abundant coarse-grained chalcopyrite, pyrite and quartz, with small amounts of sphalerite, stannite-chalcopyrite solid solution, In-bearing stannite and galena.

Under the microscope, chalcopyrite is classified in two

Table 6 Electron microprobe analysis of electrum.

Grain analyzed	Ag	wt%	Total	atom%	
		Au		Ag	Au
1-1	23.88	76.35	100.23	36.35	63.55
1-2	25.53	76.62	102.15	37.83	62.17
2-1	19.44	80.64	100.08	30.56	69.44
2-2	19.14	80.26	99.40	30.34	69.66
2-3	19.14	80.06	99.20	30.39	69.61

types by reflection color, a brassy yellow type (i.e., normal chalcopyrite) and a brownish yellow one. The latter, less common than the former, characteristically shows strong anisotropism (bluish gray to yellowish gray). This anisotropic chalcopyrite grains are rimmed by sphalerite

and In-bearing stannite. In addition, the myrmekitic intergrowth of numerous minute worm-like grains with grayish color and the size $<10\ \mu\text{m}$ and normal chalcopyrite is occasionally observed within the anisotropic chalcopyrite (Fig. 4E). As shown in Figure 5, back-scattered electron images and characteristic X-ray images of the anisotropic chalcopyrite and worm-like grains reveal that these minerals are obviously high in Sn content.

Quantitative electron microprobe analyses indicate that the Sn contents of the anisotropic chalcopyrite range from 3.3 to 4.2 wt% (Nos. 6–9 in Table 5). The chalcopyrite also contains 0.5–0.8 wt% Zn and up to about 0.1 wt% In, whereas it is depleted in Fe compared with the ideal formula. The worm-like grains contain about 0.3 wt% In and

the chemical compositions are close to $\text{Cu}_2(\text{Fe}, \text{Zn})_{1+x}(\text{Sn}, \text{In})_{1-x}\text{S}_4$ (x : about 0.25–0.45) on the basis of eight total number of atoms (Nos. 10–12 in Table 5). Therefore, the grains are regarded as stannite-chalcopyrite solid solution. On the other hand, the chalcopyrite that exhibits the myrmekitic intergrowth with the worm-like stannite-chalcopyrite solid solution is low in Sn content (0.1–0.5 wt%) and the chemical compositions correspond to the ideal formula (Nos. 13 and 14 in Table 5). The In-bearing stannite that rims the anisotropic chalcopyrite contains 3 wt% In (No. 15 in Table 5) and the chemical composition corresponds to $\text{Cu}_{1.86}(\text{Fe}, \text{Zn})_{1.22}(\text{Sn}, \text{In})_{0.93}\text{S}_{3.98}$ on the basis of eight total number of atoms. The chemical formula indicates a coupled substitution of CuSn for $(\text{Fe}^{2+},$

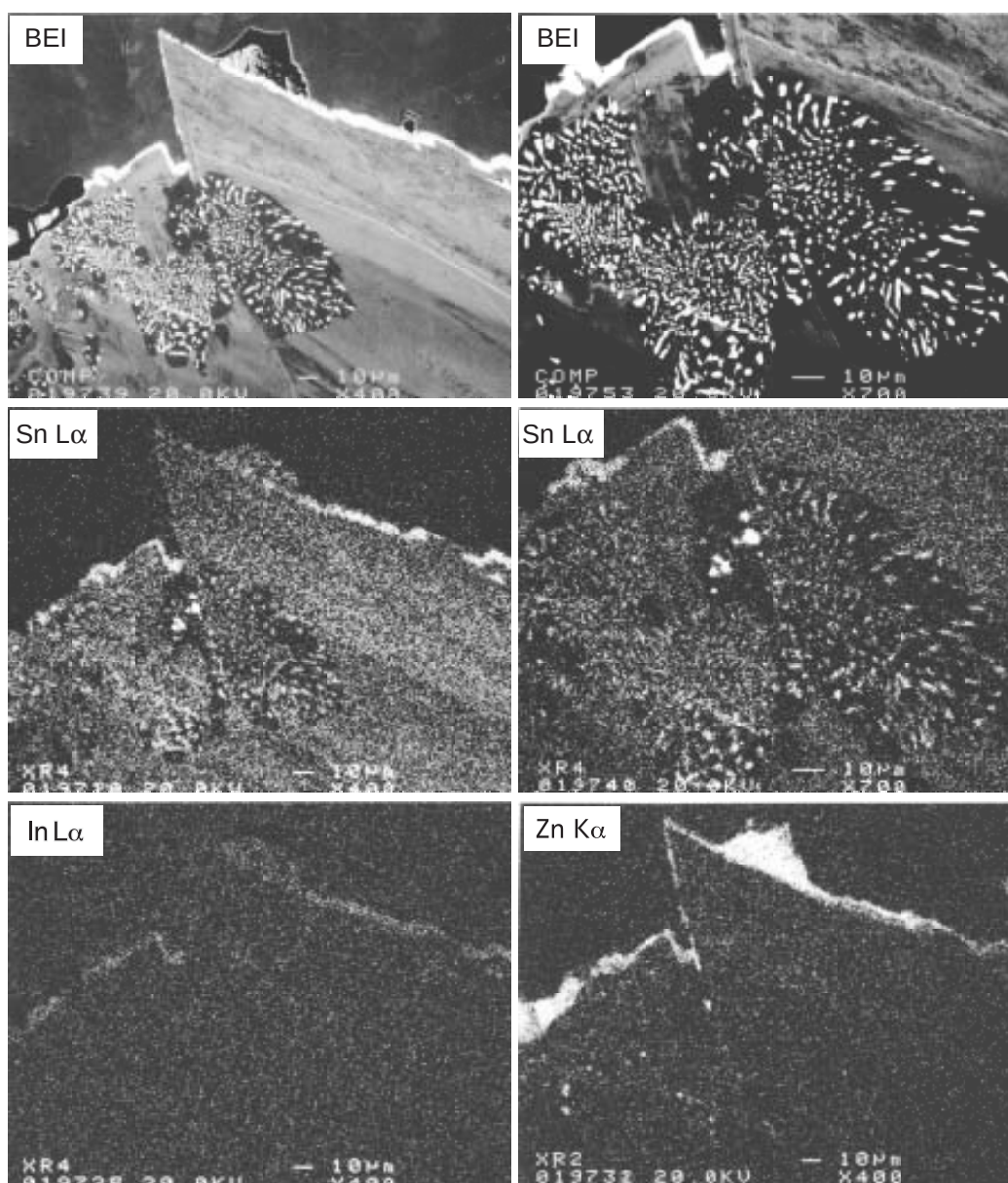


Fig. 5 Back-scattered electron images (BEI) and characteristic X-ray images of the stannite-chalcopyrite series minerals shown in Figure 4E.

ZnIn^{3+} , as recognized in stannite from Ikuno (Shimizu et al., 1986) and Toyoha (Ohta, 1989).

4.4.3. Zone III: The zone III, ranging in width from 1 to 2 cm, comprises abundant sphalerite, with subordinate amounts of chalcopyrite, galena, pyrite and marcasite pseudomorphs after pyrrhotite. Submicron-sized chalcopyrite dusts (i.e., chalcopyrite disease) develop sporadically in sphalerite. This zone is characterized by the occurrence of In- and Sn-bearing sphalerite.

The In- and Sn-bearing sphalerite (hereafter referred as In-Sn sphalerite) sporadically occurs as irregularly winding threads (1–10 μm in width) in normal sphalerite (Fig. 4F). The In-Sn sphalerite is slightly brownish in color compared with normal sphalerite and exhibits weak pleochroism and anisotropism. Characteristic X-ray images demonstrate that the In-Sn sphalerite is rich in In,

Sn and Cu and poor in Fe compared with those of the normal sphalerite (Fig. 6). Quantitative microprobe analyses show that the In, Sn and Cu contents of the In-Sn sphalerite are 2.6–8.4, 1.8–4.3 and 3.7–9.7 wt%, respectively (Table 7). By contrast with the In-Sn sphalerite, the In content is below the limit of detection, and the amounts of Sn and Cu are less than 1 wt% in the normal sphalerite. The Fe content of the In-Sn sphalerite is lower (1.7–3.4 wt%) than that of the normal sphalerite (2.5–8.6 wt%).

4.4.4. Zone IV: The zone IV is composed exclusively of coarse-grained pyrite aggregates.

5. Fluid Inclusion Microthermometry

5.1. Fluid inclusions and analytical methods

Three ore samples used for fluid inclusion microther-

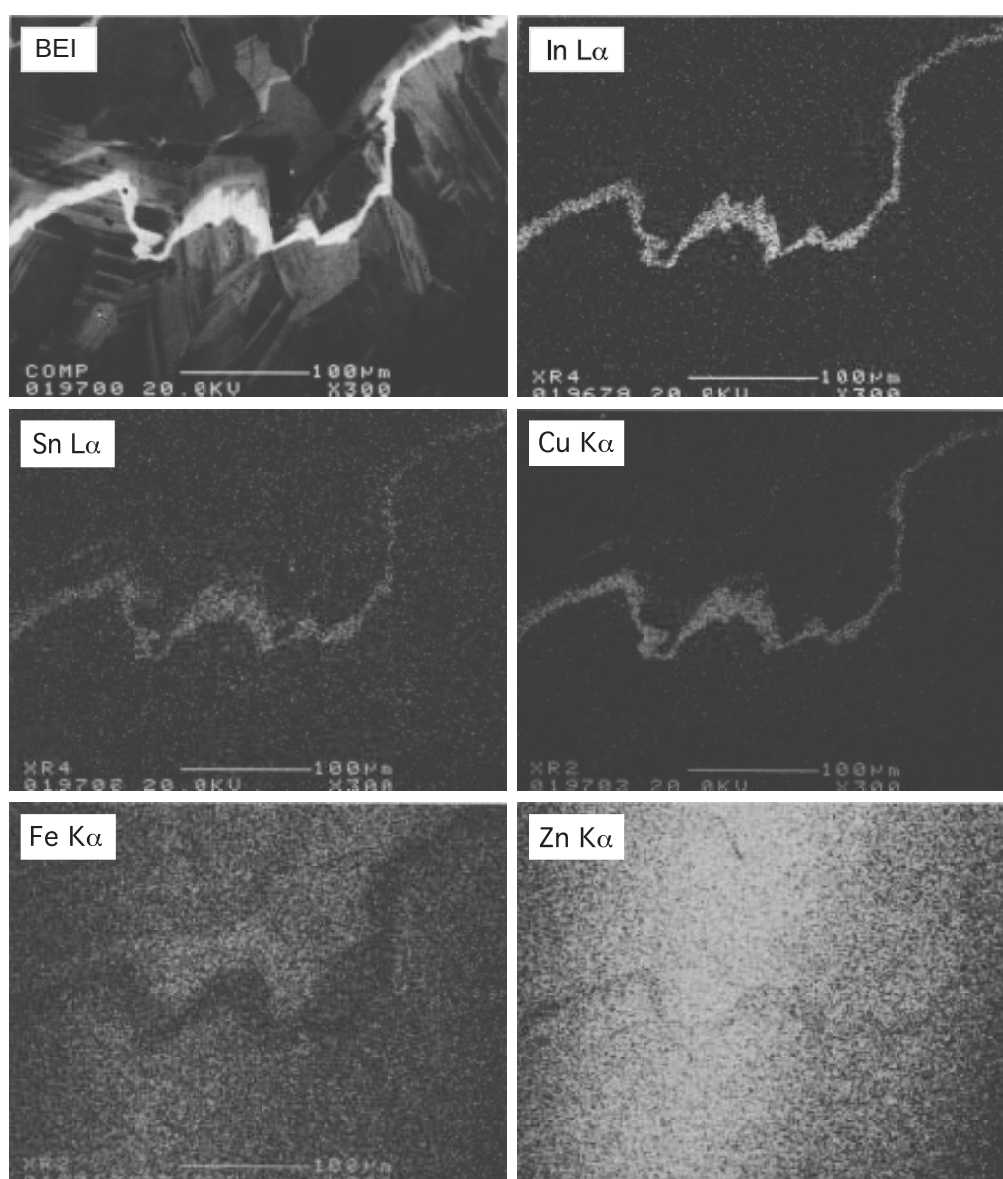


Fig. 6 Back-scattered electron image (BEI) and characteristic X-ray images of the In- and Sn-bearing sphalerite shown in Figure 4F.

Table 7 Electron microprobe analysis of sphalerite.

Mineral		Zn	Fe	Cu	Mn	Cd	In	Sn	S	Total
In- and Sn-bearing sphalerite	wt%	43.82	1.75	9.72	0.02	0.50	8.43	3.59	31.46	99.29
	atom%	34.48	1.61	7.87	0.01	0.23	3.78	1.55	50.47	100.00
	no. of atoms	2.758	0.129	0.630	0.001	0.018	0.302	0.124	4.038	8
ditto	wt%	44.78	1.94	9.10	0.02	0.50	7.84	3.45	31.41	99.04
	atom%	35.22	1.78	7.37	0.02	0.23	3.51	1.49	50.38	100.00
	no. of atoms	2.818	0.142	0.590	0.002	0.018	0.281	0.119	4.030	8
ditto	wt%	47.05	1.74	8.06	0.02	0.47	6.88	3.11	31.69	99.02
	atom%	36.78	1.59	6.48	0.02	0.21	3.06	1.34	50.52	100.00
	no. of atoms	2.942	0.127	0.518	0.002	0.017	0.245	0.107	4.042	8
ditto	wt%	45.27	2.17	8.97	0.04	0.48	6.48	4.28	31.46	99.15
	atom%	35.49	1.99	7.24	0.04	0.22	2.89	1.85	50.29	100.01
	no. of atoms	2.839	0.159	0.579	0.003	0.018	0.231	0.148	4.023	8
ditto	wt%	47.21	2.04	8.00	0.02	0.53	6.29	3.37	31.69	99.15
	atom%	36.82	1.86	6.42	0.02	0.24	2.79	1.45	50.41	100.01
	no. of atoms	2.946	0.149	0.514	0.002	0.019	0.223	0.116	4.033	8
ditto	wt%	47.43	1.78	7.95	0.02	0.45	6.43	3.24	31.86	99.16
	atom%	36.95	1.62	6.37	0.01	0.21	2.85	1.39	50.61	100.01
	no. of atoms	2.956	0.130	0.510	0.001	0.017	0.228	0.111	4.049	8
ditto	wt%	47.34	1.86	7.73	0.03	0.58	7.23	2.78	31.71	99.26
	atom%	36.94	1.69	6.21	0.02	0.26	3.21	1.19	50.46	99.98
	no. of atoms	2.955	0.135	0.497	0.002	0.021	0.257	0.095	4.037	8
ditto	wt%	54.94	3.38	3.73	0.03	0.29	2.64	1.76	32.57	99.34
	atom%	41.68	3.00	2.91	0.03	0.13	1.14	0.73	50.38	100.00
	no. of atoms	3.334	0.240	0.233	0.002	0.010	0.091	0.058	4.030	8
normal sphalerite	wt%	61.78	3.61	0.08	0.06	0.37	0.00	0.00	33.12	99.02
	atom%	46.14	3.16	0.06	0.05	0.16	0.00	0.00	50.44	100.01
	no. of atoms	0.923	0.063	0.001	0.001	0.003	0.000	0.000	1.009	2
ditto	wt%	58.29	5.80	0.59	0.42	0.01	0.00	0.48	33.32	98.91
	atom%	43.37	5.05	0.45	0.37	0.01	0.00	0.20	50.55	100.00
	no. of atoms	0.867	0.101	0.009	0.007	0.000	0.000	0.004	1.011	2
ditto	wt%	56.15	8.56	0.09	0.44	0.06	0.00	0.07	33.22	98.59
	atom%	41.72	7.45	0.07	0.39	0.03	0.00	0.03	50.32	100.01
	no. of atoms	0.834	0.149	0.001	0.008	0.001	0.000	0.001	1.006	2
ditto	wt%	61.13	2.51	0.75	0.63	0.10	0.00	0.70	33.15	98.97
	atom%	45.74	2.20	0.58	0.56	0.05	0.00	0.29	50.58	100.00
	no. of atoms	0.915	0.044	0.012	0.011	0.001	0.000	0.006	1.012	2

mometry are as follows: (1) sphalerite and quartz from the polymetallic banded ore (STT-01); (2) sphalerite and quartz from sphalerite-pyrite aggregates of the Pb-Zn-Cu ore (STT-13); and (3) sphalerite of the zone III, and quartz of the zone I from the polymetallic banded ore (STT-03) (Fig. 3). Because sphalerite in these samples is nearly opaque (black or dark red) in transmitted light in general, and transparent crystals suitable for the experiments are less common, only a few fluid inclusions in sphalerite could be examined. All the inclusions observed are generally ranging from 10 to 30 μm in size and of two-phase and liquid-rich type at room temperature. Homogenization temperatures (T_h) and temperatures of final ice-melting point (T_m) for primary fluid inclusions trapped in the minerals were measured using a Linkam TH 600PM heating/cooling stage. The origin of inclusions was distinguished on the basis of the criteria summarized by Roedder (1984). The stage was calibrated against the melting points of tin (232°C) and bismuth (271°C) and the freezing temperatures of pure water, tridecane (−5.5°C) and dodecane (−9.6°C). Uncertainties in mea-

surements of T_h and T_m are estimated to be less than $\pm 3^\circ$ and $\pm 0.1^\circ\text{C}$, respectively. Salinities (in wt% NaCl equiv.) were obtained using a table for determining the freezing point depression of H_2O -NaCl solutions given by Bodnar (1993).

Laser Raman microprobe analysis for five fluid inclusions in quartz (four from STT-01 and one from STT-13) was also performed at Fukuoka University. However, gas components such as CO_2 , N_2 , H_2S and CH_4 were not detected in these inclusions.

5.2. Results

The microthermometric results are illustrated in Figures 7 and 8. The T_h values for sphalerite in the three ore samples are ranging from 218° to 231°C, whereas those for quartz show a wider range mostly from 184° to 246°C with an exception of 295°C. The T_m and salinities of inclusion fluids trapped in sphalerite range from −2.2° to −3.2°C and from 3.7 to 5.3 wt%, while those in quartz are in the range from −1.6° to −2.3°C and from 2.7 to 3.9 wt%, respectively.

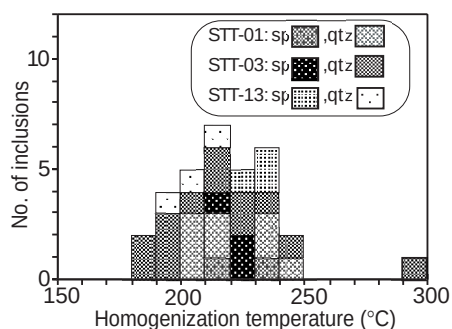


Fig. 7 Histograms of homogenization temperatures of fluid inclusions trapped in sphalerite (sp) and quartz (qtz).

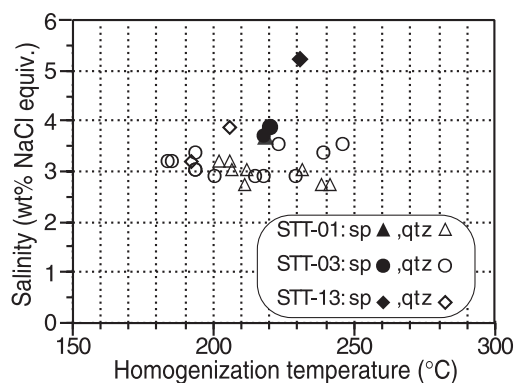


Fig. 8 Plots of salinity (wt% NaCl equiv.) vs. homogenization temperature for fluid inclusions trapped in sphalerite (sp) and quartz (qtz).

Salinities obtained from sphalerite are slightly higher than those from quartz in each ore sample.

6. Discussion

6.1. Characteristics of the polymetallic banded ore

The polymetallic banded ore from the Ohkubo vein is characterized by an abundance of Au, Ag, Sn, Bi, In, Se and Te. The results of microscopic observation and electron microprobe analysis have revealed a distinct zonal distribution of these metals in the ore: (1) Au, Ag, Sn, Bi, Se and Te (zone I); (2) Sn (and In) (zone II); and (3) Sn and In (zone III). The Th values of fluid inclusions trapped in quartz from the zone I and in sphalerite from the zone III are mostly 184°–246° and 218°–222°C, respectively. Pressure corrections for the Th of fluid inclusions estimated from geologic setting around the deposit (Suzuki et al., 1981; Yahata, 2002) and the obtained K-Ar age data (8.1–5.7 Ma) are fairly small. In contrast, most of Th of fluid inclusions and/or temperatures based on Fe and Zn partitioning between coexisting stannite and sphalerite obtained from other tin-polymetallic deposits are higher than 300°C; for example, 350°–400°C at Toyoha (Ohta, 1989, 1995), 300°–340°C at Otoge (Watanabe et al., 1994), and 178°–528°C at 22 deposits in Bolivia (Sugaki et al., 1988). In comparison with these deposits, it

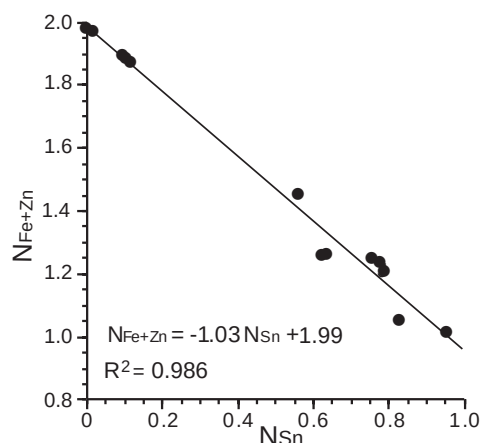


Fig. 9 Plots of the numbers of Fe+Zn ($N_{\text{Fe+Zn}}$) vs. Sn atoms (N_{Sn}) in stannite-chalcopyrite series minerals (the total number of atoms = 8). The regression line of $N_{\text{Fe+Zn}}$ on N_{Sn} is also shown. R^2 denotes a coefficient of determination.

is highly probable that the tin-polymetallic mineralization at Suttsu occurred under the conditions of low temperatures; that is, it took place at about 250°C and that the temperatures of ore fluids decreased to about 180°C with time.

One of the mineralogical features observed in the polymetallic banded ore is the occurrence of Sn-bearing anisotropic chalcopyrite (3.3–4.2 wt% Sn), which includes the myrmekitic intergrowth of stannite-chalcopyrite solid solution and normal chalcopyrite. The Sn contents of the anisotropic chalcopyrite are higher than those (up to 2.34 wt%) from Toyoha (Kase, 1987). Kase (1987) and Ohta (1989) have demonstrated a coupled substitution of $(\text{Fe}^{2+}, \text{Zn})\text{Sn}^{4+}$ for 2Fe^{3+} between normal chalcopyrite and stannite in Toyoha. The chemical compositions of the chalcopyrite-stannite solid solution series given in Table 5 (except for In-bearing stannite) indicate that the number of [Fe+Zn] atoms, calculated on the basis of eight total number of atoms per formula unit, decreases linearly with increasing number of Sn atoms, with a slope of about -1.0 (Fig. 9). Thus, the relationships also show the same substitution as that for the Toyoha specimen. An experimental study by Moh (1975), however, has revealed that chalcopyrite takes only ≤ 3 wt% stannite component below $462 \pm 5^\circ\text{C}$. Therefore, it is most likely that the Sn-bearing chalcopyrite from Suttsu, as well as from Toyoha stated by Kase (1987), has precipitated metastably. Furthermore, the mode of occurrence of the stannite-chalcopyrite solid solution and normal chalcopyrite exhibiting the myrmekitic texture and their chemical compositions suggest that the texture has resulted from the decomposition of the Sn-bearing chalcopyrite.

Another feature observed in the ore is the occurrence of In-Sn sphalerite. Electron microprobe analyses of In-Sn sphalerite (Table 7) show that the number of [In+2Sn] atoms, calculated on the basis of eight total number of

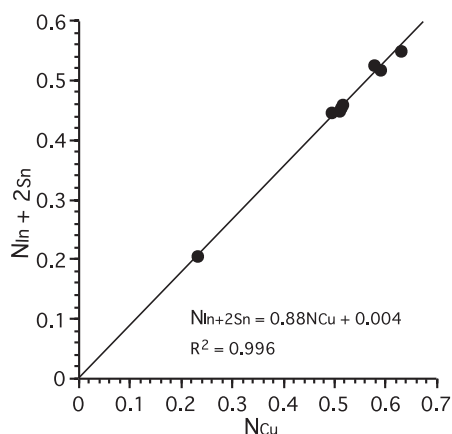


Fig. 10 Plots of the numbers of In+2Sn (N_{In+2Sn}) vs. Cu atoms (N_{Cu}) in In- and Sn-bearing sphalerite (the total number of atoms = 8). The regression line of N_{In+2Sn} on N_{Cu} is also shown. R^2 denotes a coefficient of determination.

atoms per formula unit, increases linearly with increasing number of Cu atoms, with a slope of about 0.9 (Fig. 10) and that the range of Fe/Sn atomic ratios is about 1.0–4.1 (mostly ≤ 1.4). According to Oen et al. (1980), sphalerite and stannite form limited solid solutions in the (Zn, Fe)S– Cu_2FeSnS_4 – $CuInS_2$ pseudoternary system. Therefore, it seems likely that the In–Sn sphalerite forms a solid solution of sphalerite ((Zn, Fe, Cu, Cd)S), roquesite ($CuInS_2$), and stannite (Cu_2FeSnS_4) with the contents of about 2–9 mole% roquesite and 2–4 mole% stannite.

6.2. Tin-polymetallic mineralization in southwestern Hokkaido

In addition to Suttsu, tin-polymetallic mineralization has been recognized at several ore deposits in southwestern Hokkaido, including Toyoha, Shin-Ohtoyo and Oshamanbe. Extensive Zn–Pb–Cu–Ag–Sn–In mineralization containing rare metals such as Bi, W and Co occurs in the Izumo and Shinano veins at Toyoha (Narui et al., 1988; Kanbara et al., 1989; Ohta, 1989, 1991, 1995). The K–Ar ages on sericite from the Izumo and Shinano veins are 0.49 ± 0.04 to 2.84 ± 0.27 Ma (Sawai et al., 1989; Metal Mining Agency of Japan (MMAJ), 1990) and 0.80 ± 0.06 to 2.18 ± 0.46 Ma (MMAJ, 1990), respectively. The sulfide ores from the Shin-Ohtoyo Cu–Au stockwork deposit are characterized by the occurrence of ore minerals containing Sn, V, Bi and Te (Imai et al., 1999). The K–Ar ages on sericite from the deposit are 3.10 ± 0.13 and 5.53 ± 0.19 Ma (Sawai and Itaya, 1996). The pyrite–galena banded ore from the Oshamanbe Au–Ag–Cu–Pb–Zn–Mn vein-type deposit (Matsumura, 1961) contains small amounts of pyrrhotite, sphalerite, chalcopyrite, stannite and kesterite (K. Hirai, Unpubl. data). Sericites from the argillic alteration outcrop and strongly sericitized rock occurring at the deposit yield ages of 2.8 ± 0.1 Ma (Hirai et al., 2001) and 1.69 ± 0.09 Ma

(METI, 2001), respectively. These age data and the K–Ar ages from Suttsu (5.9 ± 0.2 and 7.7 ± 0.4 Ma) therefore lend support to the statement that the tin-polymetallic mineralization in southwestern Hokkaido occurred sporadically from Late Miocene to Pleistocene.

Although ilmenite-series granitoids are one of important factors for indium mineralization (Murao et al., 1991), those of Miocene age have not been recognized around the Suttsu deposit. According to Tsukimura et al. (1987), sphalerite from vein-type deposits in Japan tends to be poor in FeS (<12 mole%) in magnetite-series granitic terranes, whereas it has high FeS content (>14 mole%) in ilmenite-series ones. However, Fe-rich sphalerite (12–19 mole% FeS) and pyrrhotite are recognized at several deposits in the former. Also in the polymetallic banded ore from Suttsu, the range of FeS content of normal sphalerite is relatively wide (6.4–15.0 mole%) and pyrrhotite occurs as a primary mineral. Oxidation and sulfidation states of the magmas genetically related to these vein-type deposits seem to be somewhat low compared with those of typical magnetite-series magmas.

Ohta (1995) has pointed out that the tin-mineral-bearing deposits in southwestern Hokkaido are spatially related to pre-Tertiary basements (mainly slate) rich in organic material and that local reduction of I-type magnetite-series magmas to the oxidation state of ilmenite-series ones is essential to cause tin-polymetallic mineralization. In Suttsu, Late Miocene quartz porphyry having a high magnetic susceptibility is emplaced to the northwest of the deposit and pre-Tertiary basement rocks composed chiefly of sandstone and slate are exposed in the area about 12 km southwest of the deposit (Suzuki et al., 1981). As already mentioned, the tin-polymetallic banded ore containing Au, Ag, Sn, Bi, In, Se and Te is only locally found in the Ohkubo vein. Namely, the occurrences of tin-bearing minerals and In–Sn sphalerite are restricted to within narrow bands of the ore. Moreover, gypsum and barite often occur in vugs of the Honpi and Ohkubo veins. These geologic and mineralogical features as well as the FeS content of sphalerite therefore suggest that the pre-Tertiary basement rocks underlie the Miocene Kunnui Formation nearby the deposit and that they contributed to local and temporary reduction of the magnetite-series magmas favorable for the early stage tin-polymetallic mineralization.

7. Summary and Conclusions

Major conclusions of this study are summarized as follows:

1. The Suttsu polymetallic mineralization occurred in Late Miocene (8.1–5.7 Ma).
2. The polymetallic banded ore found in the Ohkubo vein is characterized by an abundance of Au, Ag, Sn, Bi, In, Se and Te. These metals are mainly ascribed to electrum (30.3–37.8 atom% Ag), Se-bearing pavonite (8.5–9.5 wt% Se), gustavite–lillianite solid solution, Se-

bearing bismuthinite (5.0–5.3 wt% Se), kawazulite, cassiterite, Sn-bearing chalcopyrite (3.3–4.2 wt% Sn), In-bearing stannite, stannite-chalcopyrite solid solution, and In- and Sn-bearing sphalerite (2.6–8.4 wt% In and 1.8–4.3 wt% Sn). This sphalerite likely forms a sphalerite-roquesite-stannite solid solution with the contents of about 2–9 mole% roquesite and 2–4 mole% stannite. Temperatures and salinities (in wt% NaCl equiv.) of the ore fluids are estimated to be 180–250°C and 3–4 wt%, respectively. The Sn-bearing chalcopyrite therefore has probably precipitated metastably.

3. The geologic and mineralogical features suggest that the Miocene Kunnui Formation is underlain by the pre-Tertiary basement rocks rich in organic material nearby the deposit and that they contributed to local and temporary reduction of the magnetite-series magmas favorable for the early stage tin-polymetallic mineralization at the Suttu deposit.

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